

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

“Associate Artificial Intelligence”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

ADITYA YADAV - 1DS20AI005

Conducted at



Pegasystems

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

2023 - 2024

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “Associate Artificial Intelligence” carried out by **Aditya Yadav[1DS20AI005]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

Gowtam Karri

**Signature of Reporting
Head**
Gowtam Karri
Team Lead
Pegasystems

**Signature of Internship
Coordinator**
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
**Dr. Vindhya P
Malagi**
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **ADITYA YADAV**, final year student of Information Science and Engineering, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **PEGASYSTEMS**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date: 26/04/2026

NAME: ADITYA YADAV

Place: Bangalore

USN: 1DS20AI005

OFFER LETTER



PEGASYSTEMS WORLDWIDE INDIA PRIVATE LIMITED
Registered Office: Building No.12A, 13th Office Level,
MindSpace Cyberabad, Madhapur, Hyderabad -500081
CIN No. U72200TG2007FTC055018

Date: **17th August'2023**

Name: **Aditya Yadav**

Email ID: **yadityayadav2002@gmail.com**

Letter of Intent

Congratulations! Subsequent to the online test and the discussions you have had, we are pleased to inform you that you have been shortlisted for the position of **Associate System Architect** (Subject to change post Joining) in our organization.

In this regard, we are releasing this letter which expresses our intention to appoint you. The Final Employment Offer Letter shall be subject to the successful completion of all curricular requirements as laid down by the University/Institute for award of the degree/diploma and the prerequisites as determined by PEGASYSTEMS involving:

- Successful meeting the comprehensive documentation.
- Background verification.
- Necessary approvals.
- Contingent upon the successful completion of the requirements set within the stipulated timelines during the Internship period.

The location of placement will be based on business need (Bengaluru or Hyderabad), the Date of Joining and the other details would be communicated to you in the due course of time.

Please note, this Letter of Intent **DOES NOT** amount to a final offer or an obligation on part of PEGASYSTEMS or any of its employees, for your appointment at the company.

This LOI will be terminated subject to occurrence of any of the following –

- a) In case you are not able to provide necessary documentation as required by us
- b) If you fail to complete your curricular requirements
- c) If you fail in background verification

We are excited about the prospect of you being a part of our growing team!

Sincerely,

For Pegasystems Worldwide India Pvt. Ltd

SMRITI
Digitally signed by
SMRITI MATHUR
MATHUR
Date: 2023.08.17
19:46:24 +05'30'

Authorized Signatory

COMPLETION CERTIFICATE

RE: Request for internship certificate.



Bherwani, Dinesh

To Yadav, Aditya



Thu 11:51 AM

Start your reply all with:

[Thank you!](#)

[Received, thank you.](#)

[Thank you for your submission.](#)

[Feedback](#)

Hi Aditya,

Please find below the details requested for:

- Full Name – Aditya Yadav
- ShortID – YADAA4
- Internship Start Date – 20th February'2024
- Internship End Date – 20th June'2024
- Status – Active
- Project Details – Working in Pega Consulting Team as an Associate System Architect Intern under the Go to Market Business Unit.

Do let me know if you would require any further information or details from us.

Regards,

Dinesh

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our **Principal Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr Dinesh Bherwani**, Pegasystems, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ADITYA YADAV [1DS20AI005]

TABLE OF CONTENTS

Sl no	Description	Page no
1	INTRODUCTION	1
2	DESCRIPTION OF ORGANIZATION	2
3	SCOPE OF WORK	3
4	LEARNING OBJECTIVES AS GIVEN BY COMPANY	4
5	CHALLENGES AND SOLUTIONS	5
6	ACHIEVEMENTS AND CONTRIBUTION	6
7	REFLECTION AND ANALYSIS	7
8	RECOMMENDATIONS	8
	CONCLUSION	9
	REFERENCES	10

CHAPTER 1

INTRODUCTION

As an Associate System Architect intern at Pegasystems, I embarked on a journey of discovery, innovation, and professional growth. This internship provided me with a unique opportunity to immerse myself in the dynamic world of enterprise software development and gain invaluable hands-on experience in leveraging cutting-edge technologies to solve real-world business challenges.

Throughout my tenure, I had the privilege of working alongside seasoned professionals and talented peers, contributing to diverse projects, and honing my skills in Pega's robust suite of software solutions. This report serves as a comprehensive reflection of my internship experience, detailing the projects undertaken, lessons learned, and the impact of my contributions within the organization.

In this report, I will provide an overview of Pegasystems, elucidate the objectives and scope of my internship, discuss the methodologies employed in project execution, analyze key achievements and challenges encountered, and reflect on the lessons learned and future prospects.

Through this narrative, I aim to encapsulate the essence of my journey as an Associate System Architect intern at Pegasystems, showcasing not only the technical competencies acquired but also the personal and professional growth attained during this transformative experience.

Join me as I delve into the realm of enterprise software development, innovation, and digital transformation, as witnessed and experienced firsthand at Pegasystems. The following sections of this report will delve deeper into the details of my internship experience. We'll explore Pegasystems's role in the industry, the specific scope of my work on the e-commerce website and the learning objectives outlined by the company. I will also discuss the challenges encountered during the development process and the solutions implemented to overcome them. My achievements and contributions to the project will be highlighted, followed by reflective analysis of the skills i acquired and the areas for further development.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Founded in 1983, Pegasystems provides a powerful platform that empowers the world's leading organizations to unlock business-transforming outcomes with real-time optimization. Clients use Pega's enterprise AI decisioning and workflow automation solutions to solve their most pressing business challenges – from personalizing engagement to automating service to streamlining operations. Since its inception, Pega has built a scalable and flexible architecture to help enterprises meet today's customer demands while continuously transforming for tomorrow.

Our Service Spectrum:

At Pegasystems, our service spectrum extends far beyond traditional software offerings. We pride ourselves on delivering comprehensive solutions that transcend conventional boundaries, enabling organizations to unlock new possibilities and achieve unprecedented levels of success. Our service offerings encompass:

Enterprise Software Solutions: Pega offers a suite of robust software applications designed to address the diverse needs of modern enterprises. From customer relationship management (CRM) and business process management (BPM) to artificial intelligence (AI) and robotic process automation (RPA), our platform empowers organizations to streamline operations, enhance productivity, and drive meaningful business outcomes.

Consulting and Implementation Services: Our team of seasoned professionals and certified experts collaborate closely with clients to understand their unique requirements and tailor solutions that align with their strategic objectives.

Beyond Services: Our Core Values

At the heart of Pegasystems lie our core values, guiding principles that shape our culture, drive our actions, and define who we are as an organization. These values serve as the foundation of our success and are deeply ingrained in everything we do:

Innovation: We foster a culture of innovation, continuously pushing the boundaries

Excellence: We are committed to excellence in everything we do, striving for the highest standards of quality, performance, and customer satisfaction in all our endeavors.

Integrity: We conduct ourselves with integrity, honesty, and transparency, upholding the highest ethical standards and earning the trust and respect of our clients, partners, and colleagues.

Collaboration: We believe in the power of collaboration, working closely with clients, partners, and each other to achieve shared goals, drive innovation, and deliver exceptional results.

Customer-Centricity: We are dedicated to putting our customers first, understanding their needs, anticipating their challenges, and providing solutions that exceed their expectations and deliver tangible business value.

CHAPTER 3

SCOPE OF WORK

During my internship at Pegasystems, my primary focus was on gaining hands-on experience and proficiency in utilizing both App Studio and Dev Studio. The scope of my work encompassed various activities aimed at exploring, learning, and contributing to projects within these environments. Key aspects of my scope of work included:

App Studio Exploration and Utilization:

Familiarizing myself with the features, functionalities, and capabilities of App Studio.

Exploring the different components of App Studio, including case types, data models, user interfaces, and business rules.

Learning to design and configure applications using low-code development techniques within App Studio.

Collaborating with team members to understand project requirements and translate them into App Studio solutions.

Participating in hands-on exercises and tutorials to deepen my understanding of App Studio's capabilities and best practices.

Dev Studio Development and Customization:

Acquiring proficiency in Dev Studio, the integrated development environment (IDE) for Pega applications.

Understanding the development lifecycle and methodologies within Dev Studio, including application packaging, deployment, and testing.

Leveraging Dev Studio to customize and extend applications developed in App Studio, incorporating advanced configurations, integrations, and custom coding when necessary.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Throughout my internship at Pegasystems, the company outlined specific learning objectives to guide my professional development and ensure a meaningful and productive experience. These learning objectives were designed to align with the organization's goals and priorities, as well as my individual growth aspirations. The key learning objectives provided by the company included:

1. Mastery of App Studio and Dev Studio:

Gain proficiency in utilizing both App Studio and Dev Studio, the primary development environments for Pega applications. Understand the features, functionalities, and best practices associated with each platform, and demonstrate the ability to design, configure, and customize applications using low-code development techniques.

Understanding of Pega Platform Concepts: Develop a comprehensive understanding of the core concepts and principles underlying the Pega Platform, including case management, data modeling, business rules, and user interface design. Learn to leverage these concepts to build scalable and flexible applications that address complex business requirements.

2. Application Development Skills:

Acquire hands-on experience in designing, developing, and deploying Pega applications within real-world project scenarios. Learn to collaborate with cross-functional teams, gather requirements, and translate business needs into technical solutions using App Studio and Dev Studio.

Problem-Solving and Troubleshooting Abilities: Develop strong problem-solving and troubleshooting skills to effectively identify, analyze, and resolve issues encountered during the development lifecycle. Learn to utilize debugging tools, log analysis techniques, and collaborative problem-solving approaches to overcome challenges and ensure the successful delivery of projects.

3. Continuous Learning and Growth:

Cultivate a mindset of continuous learning and growth, actively seeking opportunities to expand knowledge, acquire new skills, and stay updated on emerging trends and technologies within the field of enterprise software development. Embrace feedback and mentorship opportunities to accelerate personal and professional development throughout the internship journey.

4. Agile Development Practices:

Gain exposure to agile development methodologies and practices commonly employed within Pega projects. Learn to adapt to changing requirements, participate in sprint planning and review sessions, and collaborate with team members in an agile development environment to deliver high-quality solutions within project timelines.

Professional Communication and Collaboration: Enhance communication and collaboration skills by actively engaging with colleagues, stakeholders, and project teams. Learn to effectively articulate ideas, provide constructive feedback, and work collaboratively towards shared goals, fostering a positive and productive working environment.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Throughout my internship at Pegasystems, I encountered several challenges related to leadership, utilization of Pega Out-of-the-Box (OOTB) features in App Studio and Dev Studio, and data modeling. These challenges presented opportunities for growth and learning, prompting me to devise effective solutions to overcome them. Below, I outline these challenges and the strategies employed to address them:

Challenge 1: Leadership Challenges: As a relatively new member of the team, I initially faced challenges in assuming leadership roles and responsibilities within project teams. I struggled to assert myself and communicate effectively in group settings, hindering my ability to contribute meaningfully to project discussions and decision-making processes.

Solution: To overcome this challenge, I actively sought opportunities to observe and learn from experienced leaders within the organization. I volunteered to take on small leadership roles within project teams, such as facilitating meetings or coordinating tasks. By gradually stepping out of my comfort zone and seeking feedback from mentors and colleagues, I was able to develop confidence in my leadership abilities and contribute more effectively to team dynamics.

Challenge 2: Utilization of Pega Out-of-the-Box Features: Challenge: While working on projects in App Studio and Dev Studio, I encountered difficulties in effectively leveraging Pega's Out-of-the-Box features to meet project requirements. I struggled to understand the full extent of OOTB capabilities available within the platform and often resorted to custom coding solutions, which led to inefficiencies and increased complexity in application development.

Solution: To address this challenge, I proactively engaged in self-directed learning and exploration of Pega's documentation, training resources, and community forums. I attended workshops and webinars focused on maximizing the use of OOTB features in application design and development. Additionally, I sought guidance from senior developers and solution architects within the organization to gain insights into best practices and techniques for leveraging OOTB functionality effectively. Through continuous experimentation and practice, I gradually enhanced my proficiency in utilizing Pega's built-in features to streamline development processes and deliver more robust solutions.

Challenge 3 : Data Modeling Challenges: Data modeling emerged as a significant challenge during the course of my internship, particularly in the context of designing complex data structures and relationships within Pega applications. I struggled to conceptualize and implement optimal data models that aligned with project requirements and scalability considerations.

Solution: To tackle this challenge, I invested time in studying data modeling principles and methodologies relevant to Pega application development. I collaborated closely with data architects and domain experts to gain insights into data modeling best practices and techniques. Additionally, I utilized resources such as Pega's Data Modeler tool and documentation to assist in visualizing and refining data models within App Studio and Dev Studio. Through iterative refinement and validation of data models against project requirements, I was able to overcome data modeling challenges and ensure the integrity and efficiency.

CHAPTER 6

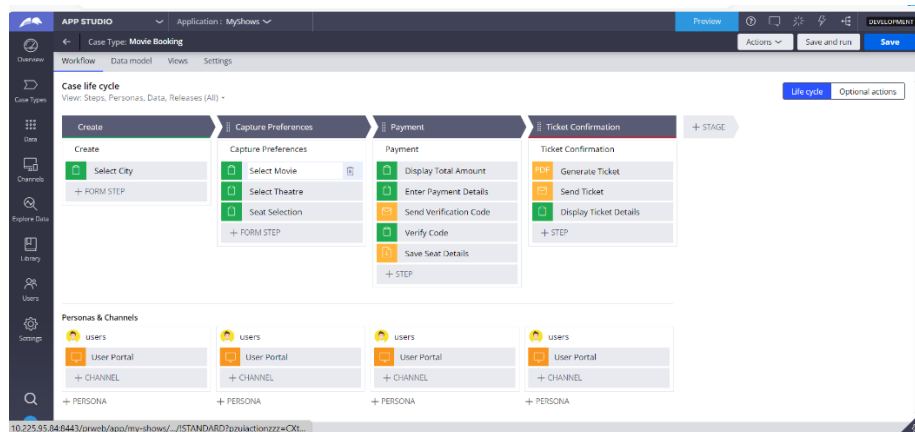
ACHIEVEMENTS AND CONTRIBUTION

During my internship at Pegasystems, I had the opportunity to contribute to the development of a comprehensive movie booking application, demonstrating my skills in application design, development, and deployment.

The following are the key achievements and contributions made during this endeavor.

Implementation of Application Life Cycle:

Orchestrated the entire application life cycle, including development, testing, deployment, and maintenance phases. Implemented agile development practices to ensure iterative and incremental delivery of application features, conducting regular sprint planning and review sessions with the project team. Leveraged version control and continuous integration tools within Dev Studio to manage code repositories and automate build and deployment processes,



ensuring seamless collaboration and integration across development environments.

Data Modeling and Database Management:

Utilized Pega's Data Modeler tool within Dev Studio to visualize and refine data structures, ensuring optimal performance, scalability, and data integrity within the application.

Implemented database queries and optimizations to enhance data retrieval and manipulation processes, optimizing application performance and responsiveness.

Integration and External Services:

Implemented integration points with payment gateways, email services, and third-party APIs to facilitate online payments, email notifications, and social media sharing features.

Ensured seamless communication and data exchange between the movie booking application and external systems, adhering to security and compliance standards.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship experience at Pegasystems has been both enriching and transformative, providing me with invaluable opportunities for learning, growth, and professional development. As I reflect on my journey, I am compelled to analyze the challenges overcome, achievements attained, and the broader implications of my experience.

Through hands-on experience with App Studio and Dev Studio, I have deepened my understanding of Pega's development environment and honed my technical skills in application design, development, and deployment. I have learned to leverage low-code development techniques, harness the power of Pega's Out-of-the-Box features, and optimize data models for performance and scalability.

Leadership and Collaboration: Overcoming challenges in leadership and collaboration has been a significant learning experience for me. I have learned the importance of effective communication, teamwork, and adaptability in navigating complex projects and dynamic team environments. By embracing opportunities to lead and collaborate with colleagues, I have cultivated valuable interpersonal skills and gained insights into effective leadership practices.

Problem-Solving and Adaptability: Throughout my internship, I have encountered various challenges, from technical complexities to project constraints. I have learned to approach challenges with a problem-solving mindset, employing critical thinking, creativity, and resourcefulness to devise effective solutions. Additionally, I have embraced adaptability, learning to thrive in fast-paced and ever-changing work environments while remaining focused on achieving project objectives.

Continuous Learning and Growth: One of the most significant takeaways from my internship is the importance of continuous learning and growth. I have embraced a growth mindset, actively seeking out opportunities for skill development, knowledge acquisition, and professional advancement. By engaging in self-directed learning, attending training sessions, and seeking mentorship from experienced professionals, I have expanded my expertise and positioned myself for future success.

CHAPTER 8

RECOMMENDATIONS

While my internship experience at Pegasystems has been largely positive, there are areas where the company could further enhance its offerings and improve the developer experience, particularly within App Studio. One recommendation is to focus on making App Studio more flexible and feature-rich, allowing developers to address a broader range of use cases without the need to switch to Dev Studio. This could involve expanding the capabilities of App Studio to support advanced customization, integration, and data modeling functionalities, empowering developers to build more complex applications entirely within the low-code environment. Additionally, providing comprehensive documentation, tutorials, and training resources specifically tailored to App Studio users would facilitate the adoption of these advanced features and promote greater self-sufficiency among developers. By investing in the evolution of App Studio as a powerful and versatile development tool, Pegasystems can streamline the development process, reduce reliance on Dev Studio, and empower developers to deliver innovative solutions more efficiently.

CHAPTER 9

CONCLUSION

My internship at Pegasystems has been a transformative journey, filled with valuable lessons, challenges overcome, and significant contributions made. Through hands-on experience with App Studio and Dev Studio, I deepened my technical proficiency in application design and development while learning to leverage Pega's Out-of-the-Box features and optimize data models. Overcoming challenges in leadership and collaboration taught me the importance of effective communication and adaptability in navigating complex projects. I embraced a problem-solving mindset and continuous learning, contributing to the development of a movie booking application and ensuring its sustainability through documentation and knowledge-sharing. As I transition from my internship role, I am inspired to continue my journey of growth and pursue new opportunities in enterprise software development, grateful for the mentorship and support provided by Pegasystems.

CHAPTER 10

REFERENCES

- How to use App studio and dev studio in Pega.
- YouTube tutorials by Pegahut.
- YouTube tutorials on Dev studio
- Pega documentation.
- Low code app builder course on Pega Academy.
- System Architect course on Pega Academy.
- Senior System Architect course on Pega Academy.
- Pega Academy GenAI course.
- Pega GenAI Blueprint.
- Creating API and Integrating Using Service REST.